

类图与设计模式说明

已根据当前 dev 分支代码更新: 新增 *Notification*、*TaskFavorite*、*UserVerification*、*TaskProgress*、*SearchController*、*FileController*、*NotificationController*、*DashboardController*。

1. 设计范围

本类图覆盖 P0 优先级能力，并与当前项目后端结构保持一致：

- 认证与用户
- 任务发布、浏览、接单、完成
- 评论、点赞、消息
- 评价与反馈仲裁
- 后台审核与用户状态管理

2. 核心类图

```
classDiagram
    class AuthController {
        +register(RegisterRequest)
        +login(LoginRequest)
        +refreshToken(String)
    }

    class UserController {
        +getCurrentUser()
        +updateSettings(UserSettingDTO)
    }

    class TaskController {
        +listTasks(TaskQuery)
        +createTask(TaskCreateRequest)
        +getTask(Long)
        +acceptTask(Long)
        +completeTask(Long)
        +toggleLike(Long)
    }

    class TaskCommentController {
        +listComments(Long, PageQuery)
        +createComment(Long, TaskCommentCreateRequest)
        +toggleCommentLike(Long)
    }

    class TaskReviewController {
```

```
        +createReview(Long, TaskReviewCreateRequest)
    }

    class MessageController {
        +listMessages(PageQuery)
        +createMessage(MessageRequest)
        +markRead(Long)
    }

    class FeedbackController {
        +createFeedback(FeedbackCreateRequest)
    }

    class NotificationController {
        +listNotifications()
        +getUnreadCount()
        +markRead(Long)
        +markAllRead()
    }

    class SearchController {
        +search(String, String, int, int)
    }

    class FileController {
        +uploadFile(MultipartFile)
    }

    class DashboardController {
        +getDashboard()
    }

    class AdminController {
        +reviewTask(Long, AdminTaskReviewRequest)
        +updateUserStatus(Long, AdminUserStatusUpdateRequest)
        +handleFeedback(Long, AdminFeedbackUpdateRequest)
        +createAnnouncement(AdminAnnouncementRequest)
        +reviewVerification(Long, AdminVerificationReviewRequest)
        +revokeVerification(Long)
    }

    class AuthService {
        +register()
        +login()
        +refresh()
    }
```

```
class UserService {  
    +getCurrentProfile()  
    +updateSettings()  
    +updateCreditScore()  
    +addPointsRecord()  
}
```

```
class TaskService {  
    +createTask()  
    +listTasks()  
    +acceptTask()  
    +completeTask()  
    +toggleLike()  
    +changeTaskStatus()  
}
```

```
class TaskCommentService {  
    +listComments()  
    +createComment()  
    +toggleLike()  
}
```

```
class TaskReviewService {  
    +createReview()  
    +aggregateResult()  
}
```

```
class MessageService {  
    +listMessages()  
    +sendMessage()  
    +markRead()  
}
```

```
class NotificationService {  
    +createNotification()  
    +getNotifications()  
    +getUnreadCount()  
    +markAsRead()  
    +markAllRead()  
}
```

```
class VerificationService {  
    +submit()  
    +review()  
    +revoke()  
}
```

```
}

class FeedbackService {
    +createFeedback()
    +handleFeedback()
}

class AdminService {
    +reviewTask()
    +updateUserStatus()
    +createAnnouncement()
}

class User {
    +id: Long
    +studentId: String
    +name: String
    +email: String
    +password: String
    +role: String
    +status: String
    +score: Decimal
    +points: Integer
}

class UserSetting {
    +id: Long
    +userId: Long
    +notificationEnabled: Boolean
    +theme: String
    +language: String
}

class Task {
    +id: Long
    +requesterId: Long
    +title: String
    +description: String
    +category: String
    +taskMode: String
    +contactInfo: String
    +reviewStatus: String
    +status: String
    +likeCount: Integer
    +expiresAt: DateTime
}
```

```
class TaskParticipant {  
    +id: Long  
    +taskId: Long  
    +participantId: Long  
    +role: String  
    +status: String  
}
```

```
class TaskComment {  
    +id: Long  
    +taskId: Long  
    +authorId: Long  
    +parentId: Long  
    +content: String  
    +likeCount: Integer  
}
```

```
class TaskLike {  
    +id: Long  
    +taskId: Long  
    +userId: Long  
}
```

```
class TaskReview {  
    +id: Long  
    +taskId: Long  
    +reviewerId: Long  
    +revieweeId: Long  
    +reviewerRole: String  
    +rating: Integer  
    +content: String  
}
```

```
class Message {  
    +id: Long  
    +senderId: Long  
    +receiverId: Long  
    +taskId: Long  
    +content: String  
    +status: String  
}
```

```
class Feedback {  
    +id: Long  
    +userId: Long
```

```
    +type: String
    +title: String
    +content: String
    +status: String
}
```

```
class Notification {
    +id: Long
    +userId: Long
    +type: String
    +title: String
    +content: String
    +referenceType: String
    +referenceId: Long
    +isRead: Boolean
}
```

```
class TaskFavorite {
    +id: Long
    +userId: Long
    +taskId: Long
}
```

```
class UserVerification {
    +id: Long
    +userId: Long
    +type: String
    +realName: String
    +studentId: String
    +imageUrls: String
    +status: String
}
```

```
class Announcement {
    +id: Long
    +authorId: Long
    +title: String
    +content: String
    +pinned: Boolean
}
```

```
AuthController --> AuthService
UserController --> UserService
TaskController --> TaskService
TaskCommentController --> TaskCommentService
TaskReviewController --> TaskReviewService
```

MessageController --> MessageService
 FeedbackController --> FeedbackService
 NotificationController --> NotificationService
 SearchController --> TaskService
 FileController --> FileService
 DashboardController --> DashboardService
 AdminController --> AdminService

AuthService --> User
 UserService --> User
 UserService --> UserSetting
 TaskService --> Task
 TaskService --> TaskParticipant
 TaskService --> TaskLike
 TaskCommentService --> TaskComment
 TaskReviewService --> TaskReview
 TaskReviewService --> Task
 TaskReviewService --> User
 MessageService --> Message
 FeedbackService --> Feedback
 AdminService --> Task
 AdminService --> Feedback
 AdminService --> Announcement

User "1" --> "1" UserSetting : owns
 User "1" --> "0..*" Task : publishes
 Task "1" --> "0..*" TaskParticipant : contains
 Task "1" --> "0..*" TaskComment : contains
 Task "1" --> "0..*" TaskLike : contains
 Task "1" --> "0..*" TaskReview : contains
 Task "1" --> "0..*" Message : context
 User "1" --> "0..*" Feedback : submits



3. 类职责说明

类/模块	职责
AuthController / AuthService	处理注册、登录、令牌刷新、登录日志记录
UserService	管理个人资料、设置、积分、信用分等用户域逻辑
TaskService	处理任务/话题发布、筛选、接单、完结、点赞、状态流转
TaskCommentService	处理评论树、回复、评论点赞
TaskReviewService	双向评价、评分汇总、成功/仲裁判定触发
MessageService	站内信收发、已读管理、任务状态提醒
FeedbackService	用户反馈、纠纷提交、处理结果维护
AdminService	后台审核、封禁、公告与仲裁处理

4. 设计模式应用

4.1 策略模式：任务模式解析与展示策略

应用位置

- TaskModeResolver
- TaskService 中对 task 与 topic 两类数据的行为分流

为什么使用

- 当前系统把“跑腿/交易”与“树洞/搭子”统一存放于 tasks 表，但业务规则不同。
- 通过策略模式可按 taskMode 分离校验、展示和状态流转规则。

如果不用会怎样

- TaskService 会堆满 if-else。
- 后续新增“咨询类”或“活动组队类”时会频繁修改旧逻辑，破坏开闭原则。

4.2 适配器模式：外部认证/内容审核/支付接入

应用位置

- P2 已定义“外部系统适配层”
- 适用于校园认证、内容审核、支付等三方能力

为什么使用

- 业务层只依赖统一接口，不直接耦合第三方 SDK。
- 便于在开发环境使用 Mock，在生产环境替换真实实现。

如果不用会怎样

- 控制器或 Service 会直接依赖第三方接口细节。
- 一旦供应商切换，改动范围会扩散到认证、任务发布、安全审核等多个模块。

4.3 模板方法模式：后台审核流程

应用位置

- `AdminService` 的任务审核、反馈处理、用户封禁

为什么使用

- 不同审核对象都遵循“加载对象 -> 校验权限 -> 更新状态 -> 记录操作 -> 发送通知”的固定流程。
- 只有状态字段和结果消息不同，适合抽公共骨架。

如果不用会怎样

- 多个后台接口会出现大量重复代码。
- 审计记录、权限校验和消息发送容易漏掉，导致行为不一致。

5. 设计结论

- 当前类划分已经具备典型的 Controller / Service / Entity 分层。
- `Task` 作为核心聚合根，连接任务、参与者、评论、点赞、评价、消息等对象。
- 为控制复杂度，建议继续把“状态流转规则”和“外部依赖”从 `TaskService` 中抽离成策略与适配接口。